Student Performance Prediction Using Machine Learning

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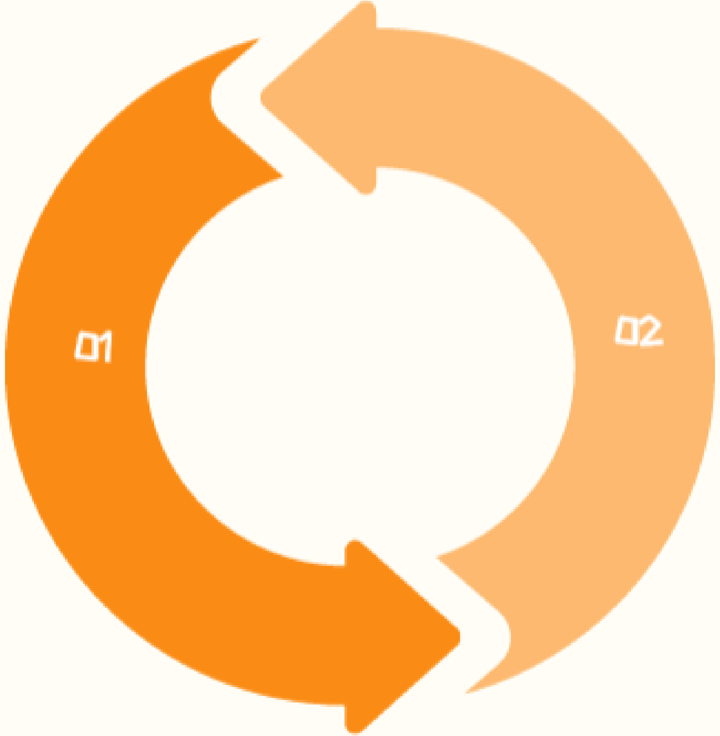
PROBLEM STATEMENT & OBJECTIVE

Student academic performance is an important indicator of educational success. Predicting student performance manually is difficult because it depends on multiple factors such as study hours, attendance, previous scores, parental involvement, and lifestyle habits.

This project develops a Machine Learning-based Student Performance Prediction System that predicts whether a student is likely to achieve good academic performance based on various input features.

Key Challenges

- Manual performance analysis is time-consuming.
- Multiple factors influence student performance.
- Difficult to identify at-risk students early.
- Educational institutions need data-driven decision making.
- Early prediction helps improve student outcomes.



INSTITUTION, ACADEMIC YEAR & MOTIVATION

UNDERTAKEN AT UNITED COLLEGE OF ENGINEERING AND RESEARCH DURING ACADEMIC YEAR 2025–2026, THIS PROJECT EMPHASIZES HOW EARLY PERFORMANCE PREDICTION SUPPORTS TIMELY INTERVENTIONS, PERSONALIZED LEARNING, AND IMPROVED EDUCATIONAL OUTCOMES.

OBJECTIVE

The primary objective is to develop a Machine Learning model that accurately predicts student academic performance.

Specific Objectives

- Collect and preprocess student performance data.
- Analyze important factors affecting performance.
- Train multiple Machine Learning models.
- Compare model performance.
- Select the best-performing model.
- Build an interactive prediction web application using Streamlit.

DATASET DESCRIPTION

THE DATASET CONTAINS ACADEMIC, PERSONAL, AND LIFESTYLE INFORMATION OF STUDENTS.

Feature	Description	Potential Influence on Performance
Hours Studied	Average daily study hours	More study time typically improves understanding and scores
Attendance	Percentage of classes attended	Higher attendance boosts exposure to teaching and materials
Previous Scores	Marks obtained in earlier exams	Strong prior scores signal foundational knowledge strength
Sleep Hours	Average sleep duration per day	Adequate sleep supports concentration and memory retention
Sample Question Papers Practiced	Number of practice exam papers solved	Practice enhances exam familiarity and reduces test anxiety
Extracurricular Activities	Level of non-academic activity participation	Balanced activities may aid skills but can reduce study time
Internet Access	Availability of online resources for study	Access expands learning materials and self-study options
Tutoring Sessions	Count of additional coaching or support classes	Extra instruction clarifies concepts and addresses weaknesses
Family Income	Approximate socio-economic status of the household	Higher income often improves access to resources and support

METHODOLOGY

- **DATA COLLECTION & PREPROCESSING** — COLLECT STUDENT PERFORMANCE DATA FROM KAGGLE, STORE LOCALLY, THEN PREPROCESS TO REMOVE NOISE, HANDLE MISSING VALUES, AND STANDARDIZE FEATURES FOR A CLEAN, ML-READY DATASET.
- **EDA, MODELING & DEPLOYMENT** — PERFORM EDA TO UNDERSTAND DISTRIBUTIONS AND KEY DRIVERS, THEN TRAIN PREDICTIVE MODELS, EVALUATE PERFORMANCE, AND FINALLY DEPLOY A STREAMLIT WEB APP FOR REAL-TIME PREDICTIONS.

DATA PREPROCESSING

HANDLING MISSING VALUES & DATA CLEANING

DETECT MISSING PATTERNS, THEN CHOOSE IMPUTATION, REMOVAL, OR SUBSTITUTION; ELIMINATE DUPLICATE RECORDS AND CORRECT INCONSISTENT FORMATS, OUT-OF-RANGE VALUES, AND OBVIOUS DATA ENTRY ERRORS SYSTEMATICALLY.

FEATURE ENGINEERING & DATA SPLITTING

NORMALIZE OR SCALE NUMERICAL FEATURES, ENCODE CATEGORICAL VARIABLES VIA SUITABLE METHODS, THEN SPLIT DATA INTO 80% TRAINING AND 20% TESTING WITH STRATIFIED SAMPLING FOR IMBALANCED CLASSIFICATION TASKS.

EXPLORATORY DATA ANALYSIS (EDA)

DATASET OVERVIEW & STATISTICAL SUMMARY

DESCRIBED DATASET SIZE, FEATURE COUNT, AND DATA TYPES, THEN COMPUTED MEAN, MEDIAN, STANDARD DEVIATION, MINIMUM, AND MAXIMUM FOR KEY VARIABLES TO UNDERSTAND CENTRAL TENDENCY AND VARIABILITY.

VISUAL ANALYSIS & KEY INSIGHTS

USED HISTOGRAMS, BOX PLOTS, SCATTER PLOTS, AND A CORRELATION HEATMAP TO REVEAL OUTLIERS, RELATIONSHIPS, AND STRONGEST PREDICTORS, HIGHLIGHTING STUDY HOURS, PREVIOUS SCORES, AND ATTENDANCE AS CRITICAL PERFORMANCE DRIVERS.

MODELS USED

MACHINE LEARNING MODELS

LINEAR REGRESSION

- BASELINE REGRESSION MODEL
- SIMPLE AND INTERPRETABLE

DECISION TREE REGRESSOR

- LEARNS DECISION RULES
- HANDLES NONLINEAR RELATIONSHIPS

RANDOM FOREST REGRESSOR

- ENSEMBLE LEARNING METHOD
- HIGHER PREDICTION ACCURACY
- REDUCED OVERFITTING

Model Evaluation

Compared using

- MAE
- MSE
- RMSE
- R^2 Score

Best model selected for deployment.

RESULT

Model Performance

Compared different Machine Learning models.

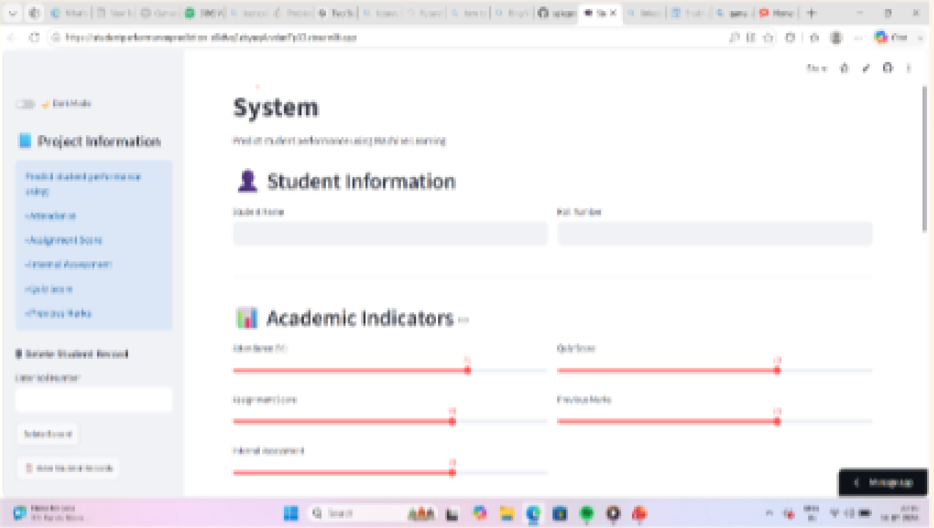
Key Findings

- Linear Regression produced baseline results.
- Decision Tree improved prediction.
- Random Forest achieved higher accuracy.
- XGBoost provided the best overall performance *(if used)*.

Final Outcome

The selected model predicts student performance accurately and can assist teachers and institutions in identifying students requiring additional academic support.

DEMO SCREENSHOTS



CONCLUSION

- SUCCESSFULLY DEVELOPED A STUDENT PERFORMANCE PREDICTION SYSTEM.
- PERFORMED PREPROCESSING AND EDA.
- COMPARED MULTIPLE MACHINE LEARNING MODELS.
- SELECTED THE BEST-PERFORMING MODEL.
- BUILT AN INTERACTIVE STREAMLIT WEB APPLICATION.
- THE MODEL HELPS PREDICT STUDENT ACADEMIC PERFORMANCE EFFECTIVELY.
- SUPPORTS TEACHERS IN IDENTIFYING WEAK STUDENTS AT AN EARLY STAGE.

FINAL OUTCOME

MACHINE LEARNING CAN IMPROVE EDUCATIONAL DECISION-MAKING BY PROVIDING ACCURATE PREDICTIONS OF STUDENT PERFORMANCE.